Right-Triangle Trigonometry

Section 1: The Six Trigonometric Ratios

1 In a right triangle with acute angle θ, the side opposite θ is 5 and the hypotenuse is 13. Find $\cos\theta$.

a. 12/13

b. 5/12

c. 13/12

d. 12/5

2 For the same triangle ($extopp = 5, \text{adj} = 12, \text{hyp} = 13$), find $\sec\theta$.

a. 13/12

b. 13/5

c. 12/13

d. 12/5

3 If $\tan\theta = 3/4$ in a right triangle, find $\csc\theta$.

a. 5/3

b. 5/4

c. 4/3

d. 3/5

Section 2: Special Right Triangles (30-60-90 and 45-45-90)

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4 In a $30^\circ - 60^\circ - 90^\circ$ triangle, if the shorter leg is 4, find the lengths of the longer leg and the hypotenuse.

a. Longer leg $= 4\sqrt{3}$, Hypotenuse $= 8$

b. Longer leg $= 8$, Hypotenuse $= 4\sqrt{3}$

c. Longer leg $= 4\sqrt{2}$, Hypotenuse $= 8$

d. Longer leg $= 6$, Hypotenuse $= 8$

5 In a $45^\circ - 45^\circ - 90^\circ$ triangle, if the hypotenuse is 10, find the length of each leg.

a. $5\sqrt{2}$

b. 5

c. $10\sqrt{2}$

d. $5\sqrt{3}$

6 What is the exact value of $\sec(60^\circ) + \tan(45^\circ)$?

a. 3

b. $2 + \sqrt{3}$

c. $1 + \sqrt{3}$

d. 2

Section 3: Cofunction Identities and Reciprocals

7 Which cofunction identity correctly relates sine and cosine for complementary acute angles?

a. $\sin(90^\circ - \theta) = \cos\theta$

b. $\sin(90^\circ - \theta) = -\cos\theta$

c. $\sin(90^\circ - \theta) = \sin\theta$

d. $\sin(90^\circ - \theta) = \tan\theta$

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8 If $\cos(28^\circ) \approx 0.8829$, what is $\sin(62^\circ)$?

a. 0.8829

b. 0.1171

c. 0.4695

d. 1.1326

9 Simplify $\cot(90^\circ - \theta) \cdot \cot\theta$.

a. 1

b. $\tan^2\theta$

c. $\cot^2\theta$

d. 0

Section 4: Angles of Elevation and Depression

10 From a point 50 ft from the base of a building, the angle of elevation to the top is 60°. How tall is the building?

a. $50\sqrt{3}$ ft

b. 100 ft

c. 50 ft

d. $25\sqrt{3}$ ft

11 A lighthouse keeper 120 ft above sea level spots a boat at an angle of depression of 30°. How far is the boat from the base of the lighthouse?

a. $120\sqrt{3}$ ft

b. $60\sqrt{3}$ ft

c. 240 ft

d. 120 ft

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12 A 20-ft ladder leans against a wall reaching a height of 16 ft. What is $\sin\theta$ where θ is the angle the ladder makes with the ground?

a. $4/5$

b. $3/5$

c. $4/3$

d. $3/4$

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